

ASSIGNMENT:2

Power Supply System Theory

Task (1):

List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

ANS.

❖ SMPS (Switched Mode Power Supply):

1. Desktop Computer:

- Converts AC power into stable DC voltages for internal computer components.

2. LED Monitor:

- Provides efficient and

regulated DC power for the display electronics.

3. Wi-Fi Router:

- Converts household AC power to the low-voltage DC required by the router.

❖ UPS (Uninterruptible Power Supply):

1) Desktop Computer:

- Keeps the computer running during a power outage, preventing data loss.

2) Wi-Fi Router/Modem:

- Maintains internet connectivity during short power cuts.

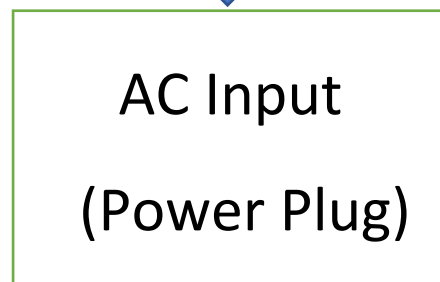
Task (2):

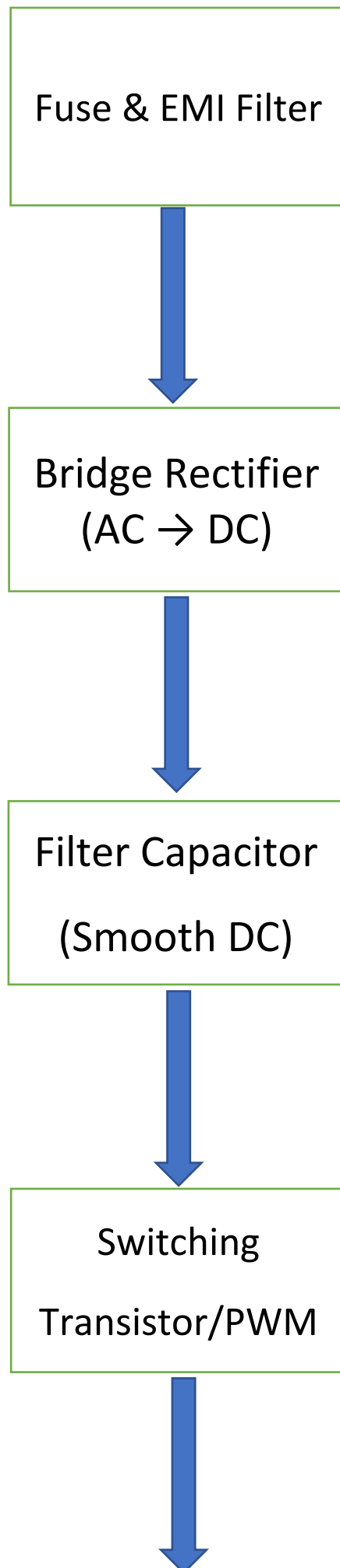
Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections. Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.

Ans.

Desktop PC SMPS (Internal Block Diagram)

230V AC Input





High-Frequency
Transformer

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graph TD; A[High-Frequency Transformer] --> B[Secondary Rectifier]; B --> C[Output Filter (Capacitors & Inductors)];
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Secondary
Rectifier

Output Filter
(Capacitors &
Inductors)

Task (3):

Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

Ans.

1. ATX 24-pin Motherboard Connector:

- Main power connector that supplies power to the motherboard.

2. SATA Power Connector:

- Used to power SATA hard drives (HDDs), solid-state drives (SSDs), and optical drives.

3. Molex (4-pin) Connector:

- Older power connector used for IDE hard drives, optical drives, case fans, and accessories.

4. PCIe 6/8-pin Connector:

- Provides additional power to dedicated graphics cards (GPUs).

Task (4)

Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Ans.

➤ Steps to Connect the SMPS in a New Gaming PC:

1. Turn off the power:

- Ensure the SMPS switch is OFF and the power cable is unplugged before making any connections.

2. Connect the motherboard power:

- Use the 24-pin ATX power connector from the SMPS.

3. Connect the CPU power

- Use the 8-pin (4+4) EPS CPU power connector.

4. Connect the graphics card (GPU)

- Install the graphics card into the PCIe x16 slot on the motherboard.

5. Connect SATA SSD/HDD

- Use the SATA power connector from the SMPS.
- Connect it to each SATA SSD or HDD.

6. Connect additional devices (if needed)

- Use Molex (4-pin) connectors for older case fans, optical drives, or accessories that require Molex power.

7. Check all connections

- Ensure every connector is fully inserted and the locking clips are

engaged.

8. Power on and test

- Connect the AC power cable to the SMPS.
 - Turn the SMPS switch ON.
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